

IDEA NAME & SUBMISSION DATE

Idea name	Nabd — an AI agent that turns coordinated mobile-network silence into a live disaster-impact map, in the first minute after the ground moves.
Submission date	13 September 2026 (Prototype Phase; Idea Phase submission shortlisted, 23 August 2026)

SUBMITTER DETAILS

Name	Kadir Can Yıldırım	Contact	kadir.can.yildirm@gmail.com · HackerEarth kadir.can.yildirm
Team	Kadir's Team — one member (solo builder)	Country	Türkiye

IDEA SUMMARY

Problem	When a disaster strikes, seismometers say <i>it happened</i> within seconds; <i>which streets, and who is cut off</i> takes hours to assemble from emergency calls — at the exact moment the network is too overloaded to carry them. In Kahramanmaraş in February 2023 that gap cost lives inside the 72-hour window. The network already held the answer: a connected block of cells that stops answering in the same second, ringed by cells that saturate as everyone still standing calls at once.
Proposed solution	A grid of sentinel SIMs (municipal or operator-owned devices, one per cell, never the public) is read every 30 s through Congestion Insights and Device Reachability Status on Nokia Network-as-Code. A LangGraph agent correlates the readings, removes what the operator's maintenance calendar explains, checks the silence against each cell's own local baseline, and declares an impact footprint only when contiguity and abnormality pass and a synchronised onset or hot ring corroborates — otherwise it abstains with the reason written down. Only inside a declared footprint, and only for people who opted in, it queries Location Retrieval to rank zones by registered people unreachable. Every pass is written to an evidence log; the command-centre console is a view over that log, never a second source of truth.
Expected benefits	An impact map 55 seconds after onset instead of hours; a ranked list of where registered vulnerable people are cut off; a false-alarm defence proven on four look-alikes and 0 of 90 ordinary mornings; privacy by construction (aggregate detection, consented triage, personal-call counts on every evidence line). Validated on two real earthquakes from USGS ShakeMaps — Kahramanmaraş 2023 and Al Haouz (Morocco) 2023 — the second on the first one's thresholds untouched. For an operator: one Open Gateway data product sold to the agency that already carries the duty of care; the bill scales with land area (≈7,800 sentinel cells for all of Türkiye), not population.

PROJECT TYPE & GSMA PILLAR ALIGNMENT

Project type	Working prototype — Python 3.12, offline-reproducible (six scenes, 121 tests, evidence byte-identical across two runners) plus a live contract run on Nokia Network-as-Code (10 September 2026, simulator devices: 18/18 calls answered, raw exchanges on file, replay parity 4/4). Public-safety / civil-protection data service; B2G via the operator.
GSMA pillar	Network intelligence — all three APIs used sit in that Open Gateway pillar (Congestion Insights, Device Reachability Status, Location Retrieval). Supports the <i>Climate Resilience & Environmental Monitoring</i> theme and the MENA smart-city / public-safety agenda; extends naturally to floods, storms and mass outages, which to the network are the same thing: an area going dark.

THEME (ONE OF 7)

Theme	6 — Climate Resilience & Environmental Monitoring
-------	---

API USAGE

CAMARA API (Nokia NaC)	What Nabd reads from it	When it is called
Congestion Insights congestion.query	Load level per sentinel cell (Low / Medium / High). A saturated ring around a silent block is the <i>hot ring</i> corroboration; congestion without silence is refused as a crowd, not an impact.	Every 30 s pass, every monitored cell (aggregate layer).
Device Reachability Status device_status.connectivity	Whether the cell's sentinel answers. A contiguous block of unreachable sentinels is the hypothesis; the local-baseline gate and the maintenance calendar decide whether that silence is abnormal <i>here</i> . Also used, with consent, to find registry members who are unreachable inside a footprint.	Every pass for sentinels; registry members only after a footprint is declared.
Location Retrieval location.retrieve	Last-seen position and uncertainty radius of an unreachable opt-in registry member, to rank zones by people cut off and give search teams a place to start.	Only inside a declared footprint, only for opted-in people, never on a quiet pass.

AI agent layer: LangGraph (Resource & Tooling Guide) — a compiled graph over a deterministic core; the personal-device node is reachable only from a verdict with an active footprint. A language model (approved providers, optional) phrases the duty officer's brief and may not invent a number: a guard rejects any figure the evidence does not carry. **Roadmap:** Geofencing subscriptions for a push-driven watch; QoD to prioritise rescue-team links inside the footprint.